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Momentum Assignment Code Summary

This project implements a momentum trading strategy in MATLAB using CRSP data from 2004–2008. The code was written to calculate monthly stock momentum, form long-short portfolios, and compute their cumulative returns. Below is a summary of how each part of the code was implemented.

1. Data Import and Setup

I began by reading the CRSP data file using readtable('crsp20042008.csv'). This loaded the data into a MATLAB table called crsp. To make time-based calculations easier, I created new variables directly within the table:

```
crsp.datenum = datenum(num2str(crsp.DateOfObservation), 'yyyymmdd');
crsp.year = year(crsp.datenum);
crsp.month = month(crsp.datenum);
```

This step converted the numeric date field into MATLAB's date format and extracted the year and month for each observation.

2. Momentum Function (getMomentum)

I defined a separate function, getMomentum, to calculate each stock's past 11-month cumulative return. Inside the function, I first computed a "multiplier date" equal to year*12 + month so I could easily look back 12 months in time.

The function filtered the main dataset to include only rows for the given PERMNO, and then located the adjusted prices for (thisMultiplierDate - 12) and (thisMultiplierDate - 1). Using those, I calculated:

```
historicReturn = (endPrice / startPrice) - 1;
```

If either price was missing or NaN, the function returned NaN. This ensured missing data did not propagate invalid results through the portfolio.

3. Applying the Momentum Function

In the main script, I looped over each observation in crsp and called getMomentum:

```
for i = 1:height(crsp)
    crsp.momentum(i) = getMomentum(crsp.PERMNO(i), crsp.year(i),
crsp.month(i), crsp);
end
```

This created a new column crsp.momentum that stored the computed momentum for every stock-month pair.

4. Portfolio Construction and Returns

To compute portfolio-level returns, I created a new table, momentum, containing all unique dates. For each month, the code:

- Subsets the CRSP data for that date.
- Uses quantile to find the 10th and 90th percentile cutoffs of the momentum distribution.
- Calculates the equal-weighted average returns for the bottom decile (mom1) and top decile (mom10):

```
momentum.mom1(k) = mean(crsp2.Returns(crsp2.momentum <= q1));
momentum.mom10(k) = mean(crsp2.Returns(crsp2.momentum >= q10));
```

When returns were missing (NaN), I explicitly set them to zero to keep arrays aligned. The long-short portfolio return was computed as momentum.mom = momentum.mom10 - momentum.mom1.

5. Cumulative Returns

Finally, I calculated cumulative performance using cumprod(momentum.momX + 1), where X is 1, 10, or the overall momentum strategy. This produced time series showing how $1 invested in each portfolio would have grown over time. This time series is also used for plotting the momentum returns.